CSC 163: Computing, Ethics, and Society

Software Law: Copyright, Licensing, and Illegal Use

ABET SO4, performance indicator 4a

Roadmap

1. **Copyright basics for software**
2. **Software licenses**: proprietary, permissive, copyleft
3. **Illegal uses of software** and who is harmed
4. **Remedies**: civil, criminal, DMCA
5. **Cases and worked examples**
6. **How to answer a 4a-style question**

Learning Objectives (SO4, PI 4a)

After this lecture you can:

- State what copyright protects in software, and what it does not
- Explain the difference between owning a copy and owning the copyright
- Compare proprietary, permissive and copyleft licenses
- Describe **two distinct situations** where using software is illegal, explain **what makes each illegal**, and identify **who is harmed**
- Name the main civil and criminal consequences

ABET SO4: recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

Copyright Basics

What the law protects when you write software

What Copyright Protects in Software

- **Protected:** the original expression in a program: source code and object code (the Copyright Office: all copyrightable expression embodied in the program)
- **Not protected:** ideas, algorithms, functions, logic, formatting and system design
- Protection is **automatic** once the code is fixed in a tangible form. Registration is optional, but a U.S. work generally must be registered before the owner can sue
- Example: anyone may write a program that uses quicksort. The algorithm is free; each program's code is protected

Source: U.S. Copyright Office, Circular 61 (computer programs)

The Owner's Exclusive Rights

Only the copyright owner (or someone with permission) may:

- **Reproduce** the program: copy it, install copies
- **Distribute** copies: sell, share, upload
- **Prepare derivative works**: modify, translate, port to another platform
- **Publicly perform or display** the work

Everyone else needs a **license** (permission), or an exception such as fair use

Owning a Copy vs Owning the Copyright

- Most software is **licensed, not sold**: you get a right to use it under stated terms
- A physical book can be resold (first-sale doctrine). Courts have often treated software as licensed, so first sale may not apply (*Vernor v. Autodesk*, 9th Cir. 2010)
- The lawful owner of a copy may make a backup copy or a copy that is an essential step in using it (17 U.S.C. 117)
- **Who owns code you write?** An employee's code within the scope of the job is usually a work made for hire, owned by the employer. Contractors and students: check the contract or university policy

Software Licenses

Permission with conditions

Types of Software Licenses

Type	Examples	What you may do	Key condition
Proprietary (EULA)	Commercial apps	Use as the license allows	No source code; no copying, redistribution or reverse engineering unless allowed
Free trial / freeware	Trials, free tiers	Use free, often limited	Not open source; may be time-limited or non-commercial
Permissive open source	MIT, Apache 2.0	Use, modify, redistribute, even in closed products	Keep copyright and license notices
Weak copyleft	LGPL	Link to the library from your own code	Changes to the library itself stay under the same terms
Strong copyleft	GPL, AGPL	Use, modify, redistribute	Larger works must be released under the same license, with source
No license	Public repo with no LICENSE file	View or fork as the host allows	All rights reserved: reuse needs the author's permission

Permissive vs Copyleft

- **Permissive (MIT, Apache 2.0):** main duty is to keep notices. Your larger work may use different terms and need not release source. Apache 2.0 adds a patent grant from contributors
- **GPLv3:** source for the licensed work and for larger works that use it must be offered under the same license
- **LGPLv3:** the library stays open; code that merely uses it through its interfaces can have other terms
- **AGPLv3:** like GPL, and also applies when a modified version runs as a **network service**
- **Key idea:** a license is a set of conditions. Breaking the conditions removes the permission

Source: choosealicense.com

Typical Restrictions in Licenses

- **Scope of use:** number of users, seats or devices; personal vs commercial; student or educational only
- **Copying and sharing:** no redistribution, no resale, no uploading to public sites
- **Reverse engineering and modification:** often banned in proprietary licenses
- **Duration and platform:** subscription term, trial period, cloud or on-premise limits
- **Attribution and share-alike:** notices to keep (permissive), source to share (copyleft)

Reading the license is a professional habit: it tells you what you may do

Illegal Uses of Software

What makes a use illegal, and who pays for it

Illegal Use: Piracy and License Violations

Situation	What makes it illegal	Who is harmed
1. Piracy: sharing or selling copies of paid software without permission	Copying and distributing without permission infringes the owner's exclusive rights (17 U.S.C. 106, 501)	Developers and publishers lose sales; people who run pirated copies face malware risk
2. Over-installing: 50 installs with 20 purchased licenses	Breach of the license terms; copies beyond the license can also infringe	The vendor loses license revenue
3. Wrong scope: using a student license for paid client work	Use outside the licensed scope: breach of contract, possibly infringement	The vendor; the program that offered the discount

Illegal Use: Circumvention, Open Source, Access

Situation	What makes it illegal	Who is harmed
4. Circumvention: keygen or patch that bypasses license checks or DRM	DMCA section 1201 bans circumventing technical measures that control access to a work	The vendor; users of cracks, who risk malware
5. Open-source violation: GPL code shipped in a closed-source product	Violating license conditions is copyright infringement (*Jacobsen v. Katzer*, Fed. Cir. 2008)	The original authors; downstream users who lose access to source
6. Unauthorized access: using a tool to log in to accounts that are not yours	Computer Fraud and Abuse Act, 18 U.S.C. 1030: access without authorization	System owners and the people whose data is exposed

Remedies and Consequences

Civil

- Court orders to stop
- Actual damages plus the infringer's profits, **or** statutory damages of **$750 to $30,000 per work**
- Up to **$150,000 per work** if willful; as low as **$200** if innocent

Criminal

- Willful infringement is a crime if done for commercial advantage or private gain
- Or if copies worth **over $1,000** are made or distributed in 180 days
- Fines and prison under federal law

Platform and contract

- DMCA takedown notices to hosts
- Account termination for repeat infringers
- Breach-of-contract claims under the license

Sources: 17 U.S.C. 504 and 506; DMCA summary (Copyright Office)

The DMCA in Brief

- **Section 1201 (anti-circumvention):** it is illegal to circumvent technical measures that control **access** to a copyrighted work, and to sell tools made for that purpose. Exceptions include security testing, encryption research and some reverse engineering
- **Section 512 (safe harbors):** online services avoid damages if they remove infringing material promptly after a proper notice
- **Notice and takedown:** owner sends a notice; the user may send a counter-notice; the material returns in 10 to 14 business days unless the owner sues
- Providers must have a **repeat-infringer** policy

Source: U.S. Copyright Office, DMCA summary

Who Is Harmed, and How

- **Developers and publishers:** lost revenue, less money for maintenance and new products
- **Open-source authors and communities:** conditions ignored, contributions taken without the promised sharing
- **Honest users:** exposure to malware in cracked software; less trust in the ecosystem
- **Society:** weaker incentives to create software, and weaker respect for rules that everyone relies on
- **Balance:** critics note that strict DRM and anti-circumvention rules can also burden legitimate users and researchers. A good answer names the harmed party **and** how they are harmed

Two Cases to Know

Vernor v. Autodesk (9th Cir. 2010)

- Vernor resold used copies of design software
- Court: buyers were **licensees**, not owners, so first sale did not apply
- Lesson: read what the license says about transfer

Jacobsen v. Katzer (Fed. Cir. 2008)

- Open-source license terms (attribution and notice of changes) were ignored
- Court: these terms are enforceable **conditions** of the copyright license
- Lesson: free software still has legal rules

Case summaries for teaching, not legal advice. Check the opinions before relying on them.

Worked Example: Applying the Three Questions

	A company installs 50 copies of a design tool, buys 20 seats	**A startup ships a GPL library inside a closed app**
What happened	Installs exceed the seats purchased	Source code of the app is not offered; GPL notice removed
Why illegal	Breach of the license scope; extra copies are not authorized	GPL conditions broken, so no permission to distribute; infringement
Who is harmed	The vendor, who loses license revenue	The library's authors, and users who cannot get the source

How to Answer a 4a-Style Question

1. Situation	Name a concrete case: piracy, over-installing, circumvention, GPL violation, unauthorized access
2. What makes it illegal	Cite the rule: copyright (copying, distribution), a license term, DMCA 1201, CFAA
3. Who is harmed	Name the party and how: lost revenue, malware, lost access to source, exposed data
4. Be distinct	A second situation must differ in kind (for example piracy vs license violation)

Level 4 answers use precise legal bases and discuss the nature of the harm.

Summary

- Copyright protects **expression** in code, automatically; ideas and algorithms are free to use
- Most software is **licensed**: conditions define what is legal
- Illegal uses include **piracy, over-installing, wrong-scope use, circumvention, open-source violations and unauthorized access**
- Consequences: injunctions, damages (statutory up to $150,000 per work if willful), criminal penalties, takedowns
- Always answer in three parts: **situation, legal basis, harmed party**

References

- U.S. Copyright Office, Circular 61: Copyright Registration of Computer Programs. copyright.gov/circs/circ61.pdf
- 17 U.S.C. chapter 5 (sections 504 and 506). copyright.gov/title17/92chap5.html
- U.S. Copyright Office, The Digital Millennium Copyright Act of 1998: summary. copyright.gov/legislation/dmca.pdf
- Choose a License: licenses overview. choosealicense.com/licenses/
- Vernor v. Autodesk, Inc., 621 F.3d 1102 (9th Cir. 2010); Jacobsen v. Katzer, 535 F.3d 1373 (Fed. Cir. 2008)
- ABET, Criteria for Accrediting Computing Programs (student outcome 4). abet.org
- These slides are original teaching summaries, not legal advice.